

Output-Driven Optimal Control of a Class of Nonlinear Systems Using Koopman Operator and High-Gain Observers

Almuatazbellah Boker¹, Khaled Aljanaideh², Michael Pumphrey³, Lamine Mili¹, Mohammad Al Janaideh³

1. Bradley Department of Electrical and Computer Engineering, Virginia Tech

2. Department of Aeronautical Engineering, University of Science and Technology

3. School of Engineering, University of Guelph

Abstract

We design an output-feedback optimal tracking controller for a class of nonlinear systems that possess full relative degree. The design procedure follows the standard LQT method using an approximate linear model of the system obtained by following the Koopman operator theory. We further identify the observables used for the Koopman method relying on output measurements only, leading to minimal data collection costs. We achieved this latter objective by realizing that output derivatives can be a good choice as observables, and hence, using high-gain observer to provide estimates of these derivatives. Overall, the proposed approach allows for solving the problem of optimal control of the considered class of nonlinear systems without the need for a prior knowledge of the system model. That is, this problem is solved and the controller is driven solely based on output measurement. We demonstrate the efficacy of the closed-loop system in controlling a power system with an infinite bus.

Keywords

control systems, high gain observers, koopman operator, optimal control, output feedback control, robotics and control systems

Output-Driven Optimal Control of a Class of Nonlinear Systems Using Koopman Operator and High-Gain Observers

Almuatazbellah M. Boker¹ Michael Pumphrey² Khaled Aljanaideh³ Lamine Mili¹ Mohammad Al Janaideh²

Abstract—We design an output-feedback optimal tracking controller for a class of nonlinear systems that possess full relative degree. The design procedure follows the standard LQT method using an approximate linear model of the system obtained by following the Koopman operator theory. We further identify the observables used for the Koopman method relying on output measurements only, leading to minimal data collection costs. We achieved this latter objective by realizing that output derivatives can be a good choice as observables, and hence, using high-gain observer to provide estimates of these derivatives. Overall, the proposed approach allows for solving the problem of optimal control of the considered class of nonlinear systems without the need for a prior knowledge of the system model. That is, this problem is solved and the controller is driven solely based on output measurement. We demonstrate the efficacy of the closed-loop system in controlling a power system with an infinite bus.

I. INTRODUCTION

Koopman operator theory is a powerful framework for modeling nonlinear dynamics by transforming them into a linear representation in a higher-dimensional space, allowing for the application of conventional control methods. It has been instrumental in addressing many challenging applications. For example, in [1] a pneumatic soft robotic arm is modeled via Koopman operator theory and was shown through experiments to predict dynamics better than other proposed methods such as neural networks. In [2] a soft robotic continuum manipulator model is identified with Koopman theory combined with model predictive control (MPC) to obtain the controller input and validated experimentally. In [3] a soft robotic Koopman operator model is trained on a computationally inefficient physics-based computer model to allow us for the control of the static shape trajectory of the flexible body. The Koopman model was then controlled via MPC which was not possible with the physics model. It was shown that the linear Koopman model was accurate and 12 times faster than the physics-based simulation. Koopman operator approach was also recently

used in [4] to model a large class of nonlinear systems to identify a non-conservative data-driven estimate of the stability region of attraction. It was also used to design a real-time Linear Quadratic Regulator (LQR) of robotic systems [5]. A key to this approach is the recognition that the derivatives of the output as the basis functions or *observables* of the Koopman operator.

In [6], a novel data-driven approach for optimal control of unknown nonlinear systems is introduced, leveraging Koopman operators and reinforcement learning techniques. This method addresses the complexities of controlling nonlinear systems by approximating them within an embedding space. This is a departure from other data-driven techniques that are based on learning the optimal controller, see for example [7], [8]. The work [9], on the other hand, proposes an end-to-end deep learning framework that jointly learns the Koopman embedding function and operator. This setup enables enhanced control by incorporating nonlinearity in the control input through an auxiliary control network, achieving notable success in complex systems such as the CartPole and robotic manipulators. Further, [10] develops a data-driven controller design for discrete-time nonlinear systems, deriving a bilinear representation with finite-gain bounds to ensure robust local stability. This combination of Koopman theory with data-driven techniques shows potential for efficiently controlling nonlinear systems in real-time applications.

Linear Quadratic Tracking (LQT) has evolved from the linear quadratic regulator to enable precise control of dynamic systems toward a specified reference trajectory, rather than simply stabilizing around a fixed point as done with LQR. This enhancement in control capability makes LQT valuable for applications where tracking a time-varying reference point is required. In [11], an inverse optimal control approach is proposed for the LQT problem, focusing on identifying unknown target states alongside weight matrices for states and inputs. The authors introduce a computationally efficient algorithm to estimate these parameters using observed optimal control inputs and state trajectories, demonstrating its application in multi-agent formation control with reduced error bounds. [12] presents an online learning algorithm designed to solve the LQT problem in continuous-time systems with partial system knowledge. By employing integral reinforcement learning, this approach finds the optimal control solution without requiring full knowledge of the system's dynamics. In [13] singular perturbation theory is used to decompose the algebraic Riccati equation (ARE) into

¹Bradley Department of Electrical and Computer Engineering, Virginia Tech, Blacksburg, VA, USA, (email: boker@vt.edu, lmili@vt.edu)

²School of Engineering, University of Guelph, Guelph, ON N1G 2W1, Canada, and Mohammad Al Janaideh is also with the Department of Mathematics, Czech Technical University, 166 36 Prague 6, Czechia (email: mpumphre@uoguelph.ca, maljanai@uoguelph.ca)

³Department of Aeronautical Engineering, Jordan University of Science and Technology, Irbid 22110, Jordan (e-mail: kfaljanaideh@just.edu.jo)

more manageable forms, enabling stable and near-optimal tracking control. These developments in LQT illustrate its flexibility and robustness for applications involving nonlinear or partially known dynamics, where adaptive tracking control is necessary.

In this work, we solve the output feedback optimal tracking control problem for nonlinear systems that have a full relative degree when the system model is unknown. In doing so, we achieve the following contributions.

- First, due to finding a Koopman approximate linear model of the system, we do not need to solve the complex Hamilton-Jacobi equation for finding the optimal controller. The achieved performance is shown to be better than the one obtained through classical linearization methods.
- Second, taking inspiration from [5], we show that, for this class of systems, the output derivatives are a good choice for the Koopman operator. These derivatives can be estimated using a high-gain observer. This is a departure from the typical way used in the literature, where the observables are chosen according to a prior knowledge of the system or based solely on guessing. Furthermore, following this technique we show that both the system modeling and control can be achieved by relying only on output measurements. This can lead to potential cost savings as it reduces the required sensors for data collection and feedback control.

The remainder of this paper is organized as follows. Section II presents the problem formulation while Section III provides background materials on Koopman theory. In Section IV, we present the main results of the paper including the how to choose the Koopman operator observables and the solution to the LQT problem. Section V provides simulation examples, and Section VI concludes with final remarks.

II. PROBLEM FORMULATION

Consider a nonlinear system of the form

$$\dot{x} = f(x, u), \quad y = Cx, \quad (1)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}$ is the control input, $y \in \mathbb{R}$ is the output, f is a sufficiently smooth (unknown) nonlinear function and C is a known constant matrix. Recall that a system's relative degree, denoted by r , is defined as the number of times the output is differentiated until an input appears. With this in mind, we have the following assumption.

Assumption 1: The system (1) has a full relative degree, that is $r = n$.

Assumption 1 essentially implies that the system does not have zero dynamics. Many applications can be modeled by (1) including the pendulum system and flexible joint manipulators [14]. The objective of the optimal tracking problem is to make the system (1) track a desired (reference) trajectory $y_d(t) \in \mathbb{R}$, generated by the command generator system

$$\dot{y}_d = F y_d, \quad (2)$$

where F is a constant matrix of appropriate dimension, while minimizing

$$J = \int_t^\infty [\gamma e^{(t-\tau)} (Cx(t) - y_d(t))^T Q (Cx(t) - y_d(t)) + u(t)^T R u(t)] dt \quad (3)$$

where $t \leq \tau$, $Q > 0$ and $R > 0$ are symmetric matrices and $\gamma > 0$ is a discount factor. It is worth emphasizing that we aim to solve this problem based only on the knowledge of the measured output y and without any prior knowledge of the system model (1). To solve this problem, we first obtain an approximate linear model of the system (1) using the Koopman operator method [15] and then follow a linear quadratic tracking (LQT) method [12] to solve for the optimal control.

III. PRELIMINARIES

A. Koopman Operator Theory

The Koopman operator theory is based on mapping the states to new higher dimensional states which are functions of the known states and are called the observable functions. Consider the nonlinear system (1). Instead of focusing directly on the state x , we examine observable functions $\psi : \mathcal{M} \rightarrow \mathbb{C}$, which are functions mapping the state space to a higher dimensional space. These observables could represent physical measurements or any quantities of interest derived from the state. The Koopman operator $\mathcal{K}$ is an infinite-dimensional linear operator that captures the time evolution of these observables. It is defined as [15]

$$\mathcal{K}\psi = \psi \circ S_t, \quad (4)$$

where $S_t : \mathcal{M} \rightarrow \mathcal{M}$ is the flow map of the system, describing how the state evolves from an initial time to time t . Essentially, applying the Koopman operator to an observable ψ yields the future value of ψ along the trajectories of the system. This approach allows us to represent the evolution of observables linearly:

$$\dot{\psi} = \mathcal{K}\psi. \quad (5)$$

Even though the original system is nonlinear, the Koopman operator acts linearly on the space of observables, which means we can utilize linear analysis techniques to study the system's behavior.

B. Extended Dynamic Mode Decomposition (EDMD)

EDMD is a data-driven approach to find the matrix $\hat{K} \in \mathbb{C}^{N_k \times N_k}$, which is a finite-dimensional approximation of the discrete-time Koopman operator $\hat{\mathcal{K}}$, for a system from sampled data of the state variables of the system [16]. From $\hat{\mathcal{K}}$, one can also obtain the Koopman eigenfunctions and Koopman eigenvalues.

In the data collection stage, the system described by (1), where f is unknown, is discretized and sampled in time Δt to create a data set of M 'snapshot pairs' (x_i, y_i) , where $x_i = x(i\Delta t)$, $y_i = y((i+1)\Delta t) = F(x_i)$ and $i = 0, \dots, M-1$. Let the data matrices be $\chi = [x_0, x_1, \dots, x_{M-1}]$ and $Y = [y_0, y_1, \dots, y_{M-1}]$, where $\chi, Y \in \mathbb{R}^{n \times M}$. In addition to

collecting the data, we create a "dictionary" of observable functions, $\Psi(x) = [\psi_1(x), \psi_2(x), \dots, \psi_{N_k}(x)]^T$, which will be used in EDMD. A further discussion of the choice of these observable functions is presented in Section IV-A. Now consider an observable function $\psi(x) \in \mathcal{F}$ that can also be written as $\psi(x) = a^T \Psi(x)$, where a is a weight vector and $\Psi(x)$ is our dictionary of observable functions. Leading to

$$\hat{K}\psi = \psi \circ F = a^T (\Psi \circ F)(x) = a^T (\hat{K}\Psi(x)) + r(x), \quad (6)$$

where $r(x)$ is a residual term. The best estimate of $\hat{K}$ takes the form

$$\hat{K} = G_1 G_0^+, \quad (7)$$

where

$$G_0 = \sum_{m=0}^{M-1} \Psi(x_m) \Psi(x_m)^T, \quad G_1 = \sum_{m=0}^{M-1} \Psi(y_m) \Psi(x_m)^T \quad (8)$$

and G_0^+ is the pseudoinverse of G_0 . Expanding to a system with input, u , the observable function list can then be adapted to also include functions of the input, or in most cases, just the input. The resulting Koopman matrix can then be decomposed into A and B matrices corresponding to the Koopman state matrix and input matrix respectively. Further information on this method is described in Section IV-B.

IV. MAIN RESULTS

A. Estimating Derivatives using High-Gain Observers

In this work, we will employ a high-gain observer to provide estimates of the derivatives of the output [17]. These estimates will be used as observables as part of the Koopman operator method as discussed in Section III to obtain an approximate linear model of the system. The estimates will also be used for the output feedback control, which is going to be designed based on the obtained Koopman model. It is worth noting that high-gain observers are essentially differentiators that enjoy robustness to modeling uncertainties and are relatively simple to design.

The EDMD method is about finding a finite-dimensional Koopman operator approximation that can accurately reflect the dynamics of the original nonlinear system. As in Section III-B, the precision of EDMD performance can vary greatly depending on the selection of observable functions $\Psi(x)$, given that those are used to determine $\hat{K}$. Essentially, selecting the best observable functions to optimize EDMD accuracy is an art form in itself. Many studies have looked at selecting the proper observable functions. For example, the work in [18] looked at using the Taylor series expansion of the system. Another work [19] showed that the optimal observable functions are the independent nonlinear terms in the system equation and the Lie derivatives of those functions. All of these previous methods have found suitable observable functions using knowledge of the system. However, in this work, we aim to solve the EDMD approximation without prior knowledge of the system model. Furthermore, we will show that for the class of considered systems, selecting the observables as the derivatives of the output provides a good

Koopman approximation and that this can be automatically done using a high-gain observer. This will be illustrated in the following example.

Academic Example

Given the nonlinear dynamical system represented as

$$\dot{x}_1 = x_2^2 - x_1, \quad (9)$$

$$\dot{x}_2 = -x_1 - x_2 - x_1^2 + u, \quad (10)$$

$$y = x_1 \quad (11)$$

Notice that the relative degree of this system is 2, which is the same as the system dimension. It is straightforward to transform the system into the chain of integrator form. This strongly suggests that a high-gain observer can be designed to provide estimates of the output and its derivatives. To this end, define $\tilde{x}_2 = x_2^2 - x_1$, then (9)-(10) becomes

$$\dot{x}_1 = \tilde{x}_2, \quad (12)$$

$$\dot{\tilde{x}}_2 = -x_1 - \tilde{x}_2 - x_1^2 + u - 2x_1(\tilde{x}_2 + x_1^2) - x_1^3, \quad (13)$$

$$y = x_1 \quad (14)$$

Then one can design a high-gain observer for the system (12)-(14) in the form [17], [20]

$$\dot{\hat{x}}_1 = \hat{x}_2 + \frac{\alpha_1}{\epsilon}(y - \hat{x}_1) \quad (15)$$

$$\dot{\hat{x}}_2 = \hat{x}_3 + \frac{\alpha_2}{\epsilon^2}(y - \hat{x}_1) \quad (16)$$

$$\vdots$$

$$\dot{\hat{x}}_N = \frac{\alpha_N}{\epsilon^N}(y - \hat{x}_1), \quad (17)$$

where $\alpha_1 - \alpha_N$ are chosen such that the polynomial $s^N + \alpha_1 s^{N-1} + \alpha_2 s^{N-2} + \dots + \alpha_N = 0$ is Hurwitz and N is the number of derivatives to be estimated. The parameter $0 < \epsilon < 1$ is chosen sufficiently small to guarantee stability and achieve desired performance. Note that $\hat{x}_2$ is the estimate of $\tilde{x}_2$ and $\hat{x}_N$ is its $(N-1)^{th}$ derivative.

Our study indicates that the larger the N the better the Koopman approximation. However, as is known, this will add the risk of making the observer more sensitive to measurement noise and may add numerical complications. Determining the optimal number of derivatives is beyond the scope of this paper and is left for future work. Nonetheless, we refer the reader to results such as in [17] that offer mitigating solutions.

B. Koopman Modeling for Control Design

If the system has no input u , and is purely autonomous, the resulting observable state matrix would be simply (5). However, if there are inputs, then the Koopman matrix needs to be split accordingly into state, A , and input, B , matrices. The dynamical system connecting the future state x_{k+1} now relies on information from both the current state x_k and the controlled input u_k , and $\Psi(x)$ becomes $\Psi(x, u)$ to incorporate the control inputs, u , as an observable state,

$\Psi_u = u$ and appended onto the observable function list [21]

$$\Psi(x, u) = \begin{bmatrix} \Psi_x(x) \\ \Psi_u(u) \end{bmatrix} = \begin{bmatrix} x \\ \psi_1(x) \\ \vdots \\ \psi_{N_o}(x) \\ u \end{bmatrix}_{N_k+l \times 1}, \quad (18)$$

where N_o is the number of state observable functions to consider and l is the number of inputs. The total size of the $\Psi_x(x)$ vector is $N_k = \aleph + N_o$. The Koopman matrix can then be separated into autonomous and driven components for the discrete case

$$\Psi_{k+1} \approx A_d \Psi_x(x_k) + B_d \Psi_u(u_k), \quad (19)$$

where A_d and B_d matrices of (19) are embedded in $\hat{K}$ and can be isolated by partitioning it as follows:

$$\hat{K} = \begin{bmatrix} A_{d_n \times n} & B_{d_n \times l} \\ \mathbf{O}_{l \times n} & \mathbf{I}_{l \times l} \end{bmatrix}, \quad (20)$$

where $A_d \in \mathbb{R}^{n \times n}$, and $B_d \in \mathbb{R}^{n \times l}$. Note $n + l = N_k$.

Since Koopman EDMD is a data-driven approach, the resulting $\hat{K}$ matrix is discrete, and hence A_d and B_d matrices. In order to convert these discrete matrices to continuous time the Tustin method is performed. The Tustin (or bilinear) transformation is commonly applied to map discrete-time systems to continuous-time representations [22]. It relates the z -domain to the s -domain via the substitution:

$$z = \frac{1 + \frac{sT}{2}}{1 - \frac{sT}{2}}, \quad (21)$$

where T is the sampling period, and s is the Laplace variable. This transformation ensures a stable mapping from the discrete to the continuous domain by approximating the derivative operator, s , with a bilinear function of the shift operator, z . Accordingly, the continuous-time state matrix A is obtained using the relation:

$$A \approx \frac{2}{T}(A_d - I)(A_d + I)^{-1}, \quad (22)$$

where I is the identity matrix. This approximation preserves system dynamics while enabling analysis and control design in the continuous domain. Similarly, the input matrix B_d is transformed as:

$$B \approx \frac{2}{T}(A_d + I)^{-1} B_d. \quad (23)$$

C. Linear Quadratic Tracking (LQT) Control

Consider the tracking control problem of the system (1) to the desired trajectory given by (2) while minimizing the control objective (3). To solve this problem, we will make use of the identified (A, B) matrices in (22) and (23) as the basis for the design of the LQT controller. We further make the following standard assumption.

Assumption 2: The pair (A, B) is stabilizable and the pair $(A, \sqrt{Q}C)$ is observable.

To solve this problem, we will follow the proposed approach in [12] and define the augmented system

$$\dot{X}(t) = \begin{bmatrix} A & 0 \\ 0 & F \end{bmatrix} X(t) + \begin{bmatrix} B \\ 0 \end{bmatrix} u(t), \quad (24)$$

where $X(t) = \begin{bmatrix} x(t) \\ y_d(t) \end{bmatrix}$.

The minimizing control input $u(t)$ is found as

$$u(t) = -R^{-1}B_1^T P X(t) \triangleq K_1 X(t), \quad (25)$$

where P is the solution the augmented Algebraic Riccati Equation (ARE):

$$0 = T^T P + P T - \gamma P - P B_1 R^{-1} B_1^T P + C_1^T Q C_1 \quad (26)$$

and $T = \begin{bmatrix} A & 0 \\ 0 & F \end{bmatrix}$, $B_1 = \begin{bmatrix} B \\ 0 \end{bmatrix}$, and $C_1 = \begin{bmatrix} C & -I \end{bmatrix}$. The discount factor γ is used to weight future errors, ensuring convergence even when the reference trajectory is not stable [12].

The output feedback control strategy is now implemented as

$$u(t) = \pm M \text{sat}\left(\frac{K_1 \hat{X}(t)}{M}\right), \quad (27)$$

where $\hat{X}(t) = \begin{bmatrix} \hat{x}(t) \\ y_d(t) \end{bmatrix}$ and $\hat{x}$ obtained from the observer (15)-(17). $\text{sat}(\cdot)$ is the saturation function that prevents the system from peaking in the observer's transient response [23]. The saturation limit M is chosen large enough to encompass the set defining the states under state feedback control. In practice, M is selected to include known limits of the system states. The structure of the closed-loop system with the proposed output feedback control is illustrated in Fig. 1.

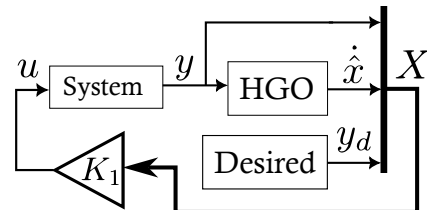


Fig. 1: The LQT controller implemented based on HGO feedback from the output of the original system.

Next, we summarize the steps involved in creating the output-driven optimal control via HGO feedback in Algorithm 1.

Algorithm 1 Output-Driven LQT Controller

Input: System output training data χ , HGO with depth N .

Output: LQT feedback gains for tracking.

- 1: *Initialisation* : Collect training data (χ) by running the system on known input (u). Record derivatives through known analytics or HGO derivatives to find $\hat{x}$. Set the observables for Koopman: $\Psi_x = \hat{x}$.
 - 2: Formulate the data matrices G_0 and G_1 via (8).
 - 3: Compute the estimated Koopman operator $\hat{K}$ via (7).
 - 4: Partition K to obtain approximated state matrix A and input matrix B via (20).
 - 5: Set LQT parameters R , Q , and γ appropriately. Solve for LQT gain matrix K_1 via (25) with appropriate saturation via (27).
 - 6: Implement the LQT controller based on Fig. 1 using feedback from desired output signal y_d and current estimates from HGO (15)-(17).
-

V. SIMULATION RESULTS

A. Academic Example Results

For the example system (12)-(14), the Koopman model is implemented with a HGO derivatives generating the observables functions in the form $\Psi_x = [x_1 \hat{x}_2 \hat{x}_3 \hat{x}_4]^T$ with an HGO derivative depth of $N = 4$. The HGO poles are set to $[-5, -6, -7, -8]$ solving the Hurwitz polynomial produces coefficients $\alpha = [26, 251, 1066, 1680]$ with an $\epsilon = 0.005$. Data is generated by injecting a sinusoidal input $u = \sin(t)$ with initial conditions $x_1(0) = 1$ and $\hat{x}_2(0) = 0$.

The A and B matrices were found to be

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -27.63 & -12.5 & -12.42 & -2.33 \end{bmatrix}$$
$$B = [0 \ 0 \ 0 \ 12.81]^T$$

The LQT controller is implemented with $R = 0.1$, $Q = 1 \times 10^8$, and $\gamma = 0.9$. This produces the following feedback gains

$$K_1 = [-30182 \quad -3121 \quad -161.38 \quad -4.8073 \quad 30185]$$

The controller is then tested against various desired trajectories as depicted in Fig. 2.

The implemented controller can be compared against a standard LQR with the gains evaluated based on linearization about the origin (equilibrium) point. A similar comparison approach is conducted in [24]. For comparison, the R and Q gains were kept the same and the cost functions are compared to show that the Koopman LQT had a lower cost for stabilization when compared to the traditional approach as shown in Fig. 3.

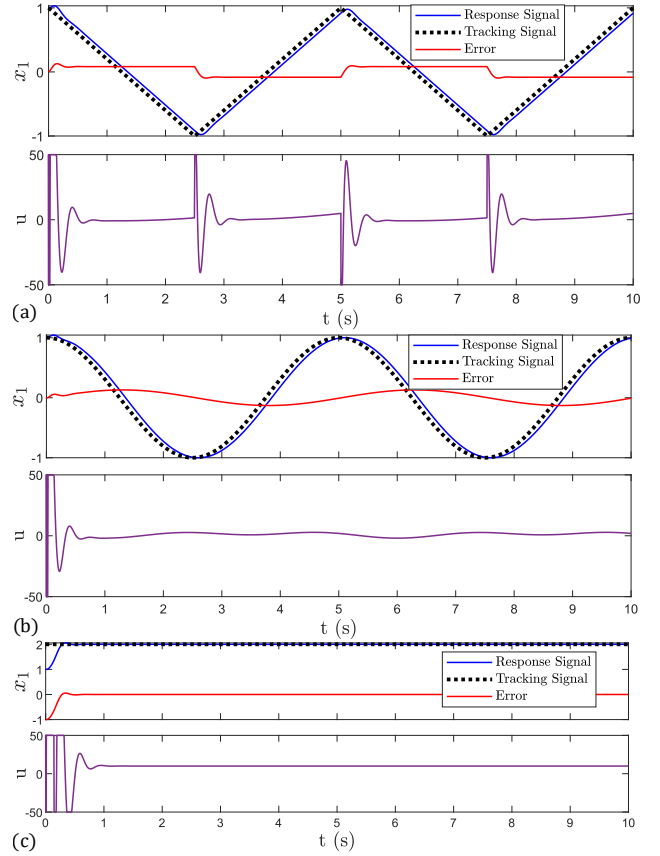


Fig. 2: The system output is shown to track the desired tracking signal when under the LQT feedback control for (a) triangular and (b) cosine signals set to a period of 5s, and (c) a step response to a value of 2, starting at 1. A saturation of the input (u) of $M = 50$ is used.

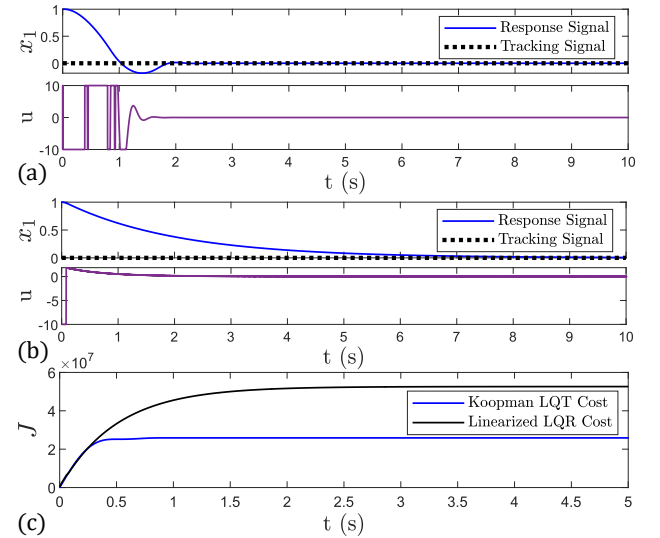


Fig. 3: The Koopman LQT (a) is set to a stabilization to $y_d = 0$. The same is done for a traditional linearized-base LQR (b). The cost functions are then compared (c) with the Koopman approach showing lower cost. A saturation of the input (u) of $M = 10$ is used.

B. Power System Example

Consider a synchronous generator connected to an infinite bus described by [25], [20]

$$\dot{\eta}_1 = \eta_2, \quad (28)$$

$$\dot{\eta}_2 = \frac{P}{M_1} - \frac{D}{M_1}\eta_2 - \frac{h_1}{M_1}\xi_1 \sin \eta_1, \quad (29)$$

$$\dot{\xi}_1 = -\frac{h_2}{\tau}\xi_1 + \frac{h_3}{\tau}\cos \eta_1 + \frac{u}{\tau}, \quad (30)$$

$$y = \eta_1 \quad (31)$$

where η_1 represents the angle in radians, η_2 denotes the angular velocity, and ξ_1 is the output voltage. In this model, P corresponds to the mechanical input power, u is the field voltage (input), D is the damping coefficient, M_1 is the inertia, and τ is the time constant. The parameters h_1 , h_2 , and h_3 are system constants. The system has a relative degree 3.

Following the steps of Algorithm 1, the high-gain observer (15)-(17) is implemented under a constant input voltage of $u = 5 \tanh t$. The HGO poles are set to $[-10, -15, -20, -25]$ solving the Hurwitz polynomial produces coefficients $\alpha = [70, 1775, 19250, 75000]$ with an $\epsilon = 0.005$. The parameters for the system is defined as follows: $P = 0$, $h_1 = 2.0$, $h_2 = 2.7$, $h_3 = 1.7$, $\tau = 6.6$, $M_1 = 0.0147$, and $D/M_1 = 4$. The initial conditions for the states are $x(0) = [1; 0.1; 1]$.

With the angular nature of the power system, it can be beneficial to add trigonometric equations to the list of observable functions [19]. For this purpose, $\sin x_1$ and $\cos x_1$ are appended to the observable function list (Ψ_x). The training data can now be generated and injected into the Koopman algorithm with the observable functions being $\Psi_x = [x_1 \ \hat{x}_2 \ \hat{x}_3 \ \hat{x}_4 \ \sin x_1 \ \cos x_1]^\top$. The produced model matches well with the original system as shown in Fig. 4.

The output Koopman A and B matrices are computed and

rounded accordingly. They are found to be

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -14.076 & -0.568 & -0.016 & 0.995 & 15.467 & -0.093 \\ -57569 & -1699.5 & -237.21 & -14.229 & 51308 & 130.23 \\ 4.013 & 0.335 & -0.018 & -0.005 & -3.951 & 0.013 \\ 1.394 & -0.631 & 0.057 & -0.007 & 11.594 & 0.623 \end{bmatrix}$$

$$B = [0 \ 0 \ 0.019 \ -26.539 \ -0.003 \ -0.126]^\top$$

Next LQT is used to develop an optimal controller given the observed states of the system. The LQT parameters used are $R = 0.1$, $Q = 1 \times 10^8$, and $\gamma = 0.1$. This produces the following feedback gains

$$K_1 = [29298 \ 2566 \ 101.22 \ 2.153 \ 1922 \ 6.922 \ -30558]$$

Through simulation the power system was controlled to follow a sinusoidal desired trajectory of $y_d = -0.5 \sin(t) + 1$, a triangle wave, and a step response to 2 as shown in Fig. 5.

Setting the feedback to just $N - 1$ derivatives was shown to work well as the last derivative is not as accurate when using HGO. A saturation is applied to the input signal u to be bounded between $\pm M$. Depending on the required tracking, the saturation may have to be increased to achieve desired response.

VI. CONCLUSION

In this work, we solve an output-feedback tracking control problem for a class of nonlinear systems that possess full relative degree. The problem is solved by minimizing a given cost function. The design procedure follows the standard LQT method made possible thanks to an approximate linear model of the system obtained by following the Koopman operator theory. We further identify the observables used for the Koopman method relying on output measurements only, leading to minimization of the costs associated with collecting system data. We achieved this latter objective by realizing that output derivatives can be a good choice as observables, and hence, we used high-gain observer to provide estimates of these derivatives. Overall, the proposed approach allows for solving the problem of optimal control of the considered class of nonlinear systems without the need for a prior knowledge of the system model. That is, this

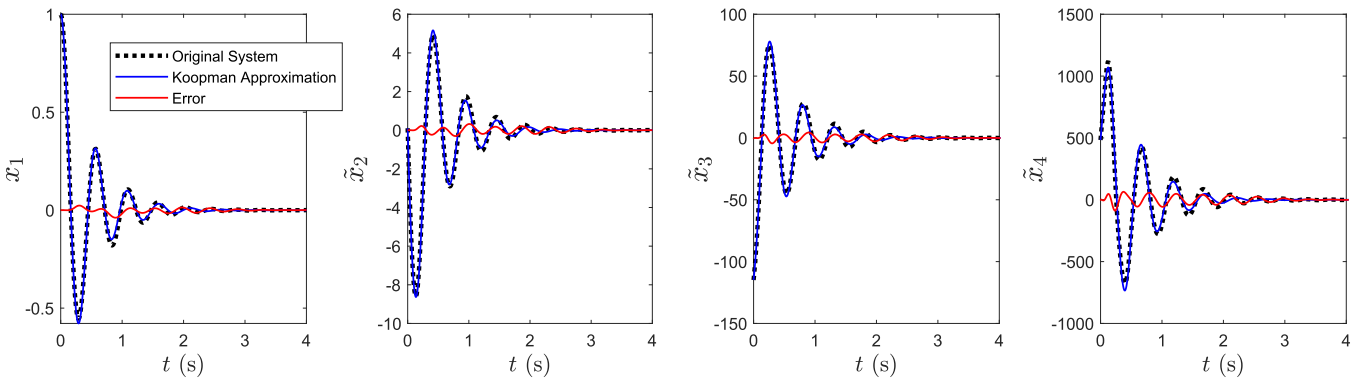


Fig. 4: The approximated Koopman model was shown to match well with the original derived power system model showing close to zero error in tracking.

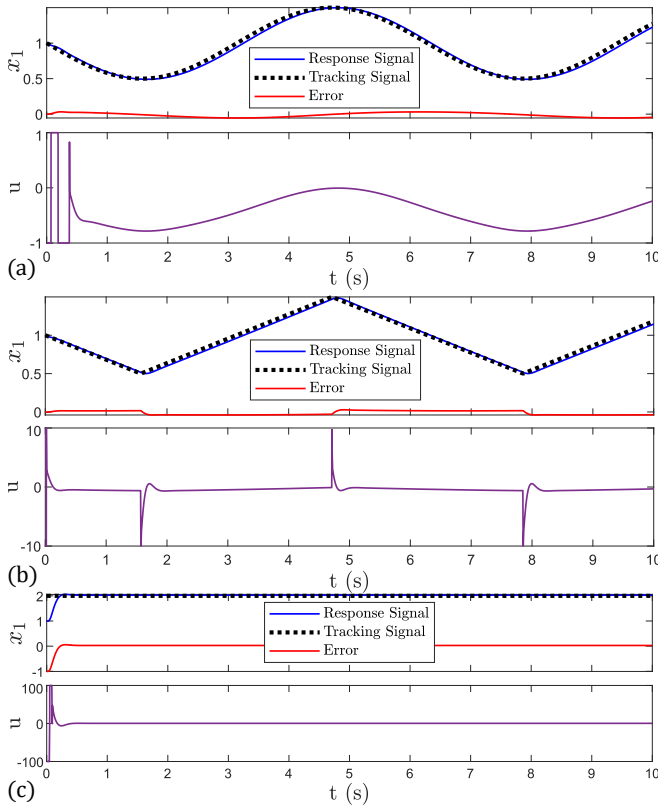


Fig. 5: The power system is controlled via LQT HGO feedback to track a desired signal of (a) $y_d = -0.5 \sin(t) + 1$, (b) $y_d = -0.5 \text{triangle}(t) + 1$ (c) $y_d = 2$. The gains for the LQT are based off the approximated Koopman model system. A saturation is applied to the input signal u to be bounded between (a) ± 1 ($M = 1$), (b) ± 10 ($M = 10$), (c) ± 100 ($M = 100$).

problem is solved and the controller is driven solely based on output measurement. This is crucial, even if the original nonlinear model is known, since solving the optimality conditions for nonlinear systems is very challenging. For future work, we aim to study the robustness of the proposed approach and provide performance guarantees.

REFERENCES

- [1] D. Bruder, C. D. Remy, and R. Vasudevan, "Nonlinear system identification of soft robot dynamics using koopman operator theory," in *2019 International Conference on Robotics and Automation (ICRA)*. IEEE, 2019, pp. 6244–6250.
- [2] H. Wang, W. Liang, B. Liang, H. Ren, Z. Du, and Y. Wu, "Robust position control of a continuum manipulator based on selective approach and koopman operator," *IEEE Transactions on Industrial Electronics*, vol. 70, no. 12, pp. 12 522–12 532, 2023.
- [3] A. Singh, J. Sun, and J. Zhao, "Controlling the shape of soft robots using the koopman operator," in *2023 American Control Conference (ACC)*, 2023, pp. 153–158.
- [4] R. Velsaco, A. Boker, R. Mili, and A. Abolmasoumi, "Data-driven estimation of region of attraction using koopman operator and reverse trajectory," in *2024 Modeling, Estimation, and Control Conference (MECC)*. ASME, 2024.
- [5] G. Mamakoukas, M. L. Castano, X. Tan, and T. D. Murphey, "Derivative-based koopman operators for real-time control of robotic systems," *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 2173–2192, 2021.

- [6] F. Zanini and A. Chiuso, "Data-driven control of nonlinear systems: Learning koopman operators for policy gradient," in *2021 60th IEEE Conference on Decision and Control (CDC)*, 2021, pp. 6491–6496.
- [7] V. Reddy, A. Boker, and H. Eldardiry, "Learning-based optimal control of linear time-varying systems over large time intervals," *Systems & Control Letters*, vol. 185, p. 105750, 2024.
- [8] V. Reddy, H. Eldardiry, and A. Boker, "Data-driven near-optimal control of nonlinear systems over finite horizon," *arXiv preprint arXiv:2306.05482*, 2023.
- [9] H. Shi and M. Q.-H. Meng, "Deep koopman operator with control for nonlinear systems," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7700–7707, 2022.
- [10] R. Strässer, J. Berberich, and F. Allgöwer, "Robust data-driven control for nonlinear systems using the koopman operator*," *IFAC-PapersOnLine*, vol. 56, no. 2, pp. 2257–2262, 2023, 22nd IFAC World Congress. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S240589632301594X>
- [11] Y. Li, C. Yu, H. Fang, and J. Chen, "Inverse optimal control for linear quadratic tracking with unknown target states," 2024.
- [12] H. Modares and F. L. Lewis, "Linear quadratic tracking control of partially-unknown continuous-time systems using reinforcement learning," *IEEE Transactions on Automatic Control*, vol. 59, no. 11, pp. 3051–3056, 2014.
- [13] J. Zhao, C. Yang, W. Gao, H. Modares, X. Chen, and W. Dai, "Linear quadratic tracking control of unknown systems: A two-phase reinforcement learning method," *Automatica*, vol. 148, p. 110761, 2023. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0005109822006276>
- [14] H. K. Khalil, "Nonlinear systems third edition," *Patience Hall*, vol. 115, 2002.
- [15] A. Mauroy, Y. Susuki, and I. Mezić, *Koopman operator in systems and control*. Springer, 2020.
- [16] M. O. Williams, I. G. Kevrekidis, and C. W. Rowley, "A data-driven approximation of the koopman operator: Extending dynamic mode decomposition," *Journal of Nonlinear Science*, vol. 25, no. 6, pp. 1307–1346, 2015.
- [17] H. K. Khalil, *High-Gain Observers in Nonlinear Feedback Control*. SIAM, 2017.
- [18] A. Mauroy and I. Mezić, "Global stability analysis using the eigenfunctions of the koopman operator," *IEEE Transactions on Automatic Control*, vol. 61, no. 11, pp. 3356–3369, 2016.
- [19] M. Netto, Y. Susuki, V. Krishnan, and Y. Zhang, "On analytical construction of observable functions in extended dynamic mode decomposition for nonlinear estimation and prediction," in *2021 American Control Conference (ACC)*. IEEE, 2021, pp. 4190–4195.
- [20] A. M. Boker and H. K. Khalil, "Nonlinear observers comprising high-gain observers and extended kalman filters," *Automatica*, vol. 49, no. 12, pp. 3583–3590, 2013. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0005109813004329>
- [21] I. Abraham, G. De La Torre, and T. D. Murphey, "Model-based control using koopman operators," in *Robotics: Science and Systems (RSS)*, vol. 13. Cambridge, Massachusetts: Robotics: Science and Systems Foundation, 2017, pp. 1–8.
- [22] B. M. Chen, *Structural Mappings of Bilinear Transformations*. London: Springer London, 2000, pp. 57–82. [Online]. Available: https://doi.org/10.1007/978-1-4471-3653-8_3
- [23] F. Esfandiari and H. K. Khalil, "Output feedback stabilization of fully linearizable systems," *International Journal of control*, vol. 56, no. 5, pp. 1007–1037, 1992.
- [24] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite linear representations of nonlinear dynamical systems for control," *PLOS ONE*, vol. 11, no. 2, pp. 1–19, 02 2016.
- [25] M. Pai, *Power system stability: analysis by the direct method of Lyapunov*. North-Holland Publishing Company, 1981.